

Healthy Living — Vision Document

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SOEN 6481 Systems Requirements Specification — Delivery #1 — Instructor: Dr. Morales

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1. Introduction

This document defines the vision for Healthy Living, a web-based application that helps students plan meals, track groceries, and find easy recipes. Its purpose is to state, at a high level of abstraction, the problem being solved, who the system serves, the capabilities it must provide, and the constraints it must respect. It describes what is needed and why, not how the system will be built; detailed behaviour will be captured in the use-case model and the Software Requirements Specification.

The software solution is a browser-based application backed by a database. It maintains a student's pantry, lets them plan meals for a day or a week, suggests recipes they can actually cook with what they already own and can afford, and turns the plan into an accurate shopping list. Around this core it provides the administration, content-moderation, and support capabilities needed to operate the platform safely.

1.1 References

- SOEN 6481 — Delivery #1 assignment description, Dr. Morales, Concordia University, 2026. Moodle course website.
- Vision Document template (Rational Unified Process artefact set), provided with the assignment.
- ISO/IEC/IEEE 29148:2018 — Requirements engineering. ISO/IEEE.
- Web Content Accessibility Guidelines (WCAG) 2.1, W3C Recommendation, 2018.
- Canada's Food Guide, Health Canada — nutrition reference source.
- Act respecting the protection of personal information in the private sector (Quebec Law 25). Publications Québec.

2. Positioning

2.1 Problem Statement

The problem of	unplanned, ad-hoc eating — deciding what to eat only when hungry, with no view of what is already in the kitchen, what it costs, or whether it is safe to eat given one's dietary restrictions
Affects	post-secondary students who prepare their own meals, and the campus health staff and families concerned with their wellbeing
The impact of which is	groceries bought and thrown away unused; money spent on expensive convenience meals; repetitive, unbalanced diets; decision fatigue; time lost to improvised grocery trips; and, for students with allergies or religious or ethical restrictions, the burden of checking every ingredient by hand
A successful solution would be	a single web application that knows what the student already has, proposes recipes that fit their budget, schedule, and dietary restrictions, turns a weekly plan into an accurate shopping list, and warns them before food goes to waste — while asking little enough effort that a busy student keeps using it

2.2 Product Position Statement

For	post-secondary students who prepare their own meals
Who	need to eat well on a limited budget and limited time, and today plan meals, track groceries, and search for recipes in three disconnected places, or not at all
The Healthy Living	is a web-based meal-planning and grocery-tracking application connected to a moderated recipe database
That	links the student's pantry, meal plan, and recipe catalogue, so that suggestions and shopping lists reflect what they actually have, what they can afford, and what they are able to eat
Unlike	recipe sites, calorie trackers, meal-kit subscriptions, and the paper lists and spreadsheets students use today, each of which solves one piece of the problem
Our product	closes the loop between inventory, plan, and purchase; is free and requires no installation; and guarantees, through active moderation, that recipes are accurate and correctly labelled for allergens

3. Stakeholder Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Project Sponsor	Funds the project and owns the business case.	Approves the vision and scope; arbitrates scope, cost, and schedule; accepts each delivery.
Requirements Engineer	Elicits, analyses, and specifies the requirements.	Produces and maintains this Vision document; validates requirements with sponsor and users.
Development Team	Designs, implements, and tests the application and database.	Delivers the features of Section 5 within the constraints of Section 6; ensures the system is maintainable.
Campus Health and Food Services	Institutional partner for student nutrition programmes.	Supplies nutrition guidance; consumes anonymised usage reporting; promotes adoption.
Data Protection Officer	Guards compliance with privacy legislation.	Approves handling of personal and health-adjacent data (allergens, restrictions); sets consent and retention policy.
Recipe Contributors	Students, dietitians, and third parties who submit recipes.	Provide the content the suggestion feature depends on, with accurate ingredient and allergen information.

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Student (Primary End-User)	The central persona for whom the app is built; uses the front end to manage their daily intake.	Maintains dietary profile and pantry; builds and follows meal plans; shops from the generated list; rates and reports recipes.	Self-represented
System Administrator (Admin)	Technical user responsible for backend maintenance, security, and overall health of the application.	Manages accounts and roles; monitors availability, performance, and security; performs backup and recovery.	Self-represented
Content Moderator	User responsible for overseeing the recipe database to ensure content quality and safety.	Reviews submitted recipes; corrects ingredient and allergen metadata; approves, rejects, or retires content.	Represents Recipe Contributors
Support Specialist	User who handles customer service and addresses technical or functional grievances.	Triages grievances; reproduces problems; resolves account issues; escalates confirmed defects.	Represents the Student in the defect process

3.3 User Environment

- Meal planning is a single-person task; the student plans, shops, and cooks alone. This is not expected to change.
- The task cycle is weekly. Planning takes five to fifteen minutes once or twice a week, usually in the evening, and is often interrupted. Pantry updates happen many times a week in bursts of seconds. Shopping happens once or twice a week for twenty to forty minutes. Time spent planning is under constant downward pressure and is the first activity dropped in a busy term.
- Environmental constraints: the shopping list is used standing in a grocery aisle, one-handed, on a phone, on a weak or absent connection. Pantry updates happen in a kitchen with busy hands. Cooking is limited to a shared kitchen, one or two burners, and about thirty minutes on a weekday.
- Platforms in use today: modern web browsers on personal laptops, shared campus machines (where students have no rights to install software), and phones. Future platforms may include native mobile applications, but the browser must remain primary.
- Other applications in use: notes apps, spreadsheets, generic recipe sites, and standalone calorie trackers. None communicate with each other; the student is the integration layer. Healthy Living does not need to integrate with them — it must replace them.
- Students receive no training, read no manual, and have no help desk; the interface must be self-explanatory on first use. The back-office users (Admin, Moderator, Support) are trained, work from desktop machines in long sessions, and can be expected to read documentation.

3.4 Key Stakeholder or User Needs

There is no existing software system; today every task below is performed by hand.

Need	Priority	Concerns	Current Solution	Proposed Solutions
Know what can be cooked now with what is on hand	High	Food already bought is wasted; the student falls back on convenience meals. No record exists of what is in the fridge.	Look in the fridge, remember, improvise; recipe sites search by name, not inventory	Pantry tracking (FE-2) feeding pantry-aware suggestion (FE-5)
Never be offered unsafe or unacceptable food	High	Allergens are a safety risk, not a preference; religious and ethical restrictions are non-negotiable.	Read every ingredient list by hand, every time	Dietary profile (FE-1) enforced as a global filter on all search and suggestion
Keep food spending within budget	High	Unplanned trips and duplicate purchases wreck a fixed budget; cost is invisible before shopping.	Mental arithmetic in the aisle; an abandoned spreadsheet	Cost shown per recipe and per plan (FE-3); budget stored in the profile constrains suggestion (FE-1, FE-5)
Reduce the time and effort planning takes	High	Decision fatigue; planning is abandoned in a busy week because it is three separate manual activities.	Paper lists, notes apps, or no plan at all	Weekly meal planner with reusable plans (FE-3)
Buy exactly what is missing	High	Forgotten items force a second trip; duplicates spoil. The list is written from memory, not derived from the plan.	A handwritten or notes-app list from memory	Shopping list generated as plan minus pantry (FE-4), with checked-off items returning to the pantry
Trust that a recipe is accurate and safe	High	Wrong allergen or ingredient data harms students; open recipe sites are unmoderated.	Cross-check recipes across several sites and hope	Moderated catalogue with a review workflow and audit trail (FE-8), fed by student reports (FE-11)
Stop groceries from spoiling unused	Medium	Money and food wasted; expiry is noticed only when it is too late.	Memory and the occasional fridge audit	Expiry tracking with use-it-first prompts (FE-6)
Know whether the week is balanced	Medium	Diets skew repetitive; nutrition data is entered by hand after the fact, if at all.	A calorie-tracking app, filled in manually, soon abandoned	Nutrition summaries derived automatically from the plan (FE-7)
Operate the platform securely and reliably	High (Admin)	Health-adjacent personal data; load peaks at evening planning times.	None — no system exists today	Administration console with accounts, monitoring, audit logs, backup and restore (FE-9)

Need	Priority	Concerns	Current Solution	Proposed Solutions
Resolve grievances before users abandon the product	Medium (Support)	An unresolved problem means a lost user.	None — no system exists today	In-app grievance raising with triage and escalation (FE-10)

4. Product Overview

4.1 Product Perspective

Healthy Living is a new, independent, and entirely self-contained web-based application. It is not a component of a larger system and does not replace an existing product: today its tasks are performed manually. It comprises a browser-based front end serving all four user classes through role-specific views, a server-side back end holding the application logic (pantry reconciliation, plan costing, shopping-list derivation, recipe ranking, moderation workflow), and a relational database storing accounts, dietary profiles, pantry inventories, meal plans, the recipe catalogue with its ingredient and allergen metadata, and moderation and support records.

Its only external interfaces are peripheral: an email service for account verification and notification, and a public nutrition reference dataset used to compute nutrition summaries. Neither sits on the critical path — if the nutrition dataset is unavailable, planning, suggestion, and shopping-list generation still work. The product does not interface with grocery-store, payment, learning-management, or wearable systems, and is not required to integrate with the tools students use today; it is intended to displace them.

4.2 Assumptions and Dependencies

Assumptions	Dependencies
Students will maintain a pantry inventory, provided each update takes seconds. This is the load-bearing assumption: if it is false, FE-4 and FE-5 lose their basis and this vision must change.	The usability of pantry entry (FE-2).
Students have regular browser access and intermittent but usable connectivity while shopping.	A mobile-usable shopping-list view that tolerates a weak connection.
A sufficient corpus of quick, low-cost, correctly labelled recipes exists at launch.	Recipe contributors, and moderation capacity to review submissions before launch (FE-8). Suggestion quality is bounded by catalogue size.
Moderation capacity scales with submission volume; opening community submission without staffing the queue breaks the safety guarantee.	Content Moderator staffing; moderation tooling with filters and batch actions.
Nutrition figures from a public dataset are accurate enough for general guidance, and students understand the product is informational.	Availability and licensing of the nutrition dataset; a visible non-medical-advice disclaimer.
Personal data, including allergens and dietary restrictions, can be stored lawfully with student consent.	Compliance with Quebec Law 25 and GDPR where applicable; approval by the Data Protection Officer.

Assumptions	Dependencies
The product remains free to students and is funded as a campus service.	Hosting costs within budget; no advertising in suggestion, since paid ranking would destroy trust in FE-5.

5. Product Features

The high-level capabilities the system must deliver. Each is externally perceivable by a user and is stated as a capability and its reason, not a design. Each will be expanded in the use-case model.

ID	Feature	Description	Priority
FE-1	Student account and dietary profile	Register, authenticate, and record allergens, dietary restrictions, disliked ingredients, weekly budget, and nutrition goals. The profile acts as a global filter: no recipe violating a declared allergen or restriction is ever presented, and the student can see why one was excluded. Needed because allergens are a safety concern and manual ingredient checking is a main reason students give up cooking.	Critical
FE-2	Grocery and pantry tracking	Record what groceries are on hand — item, quantity, unit, optional expiry — and add, consume, or discard them with minimal effort. The pantry is the system's model of the real kitchen and the input to FE-4, FE-5, and FE-6; its value collapses if it is stale, so speed of entry is the dominant concern.	Critical
FE-3	Meal planning	Assign recipes to meal slots across a day or week, adjust as the week changes, and reuse or duplicate a previous plan. Aggregate cost and nutrition are shown as the plan is built, so trade-offs are visible when the decision is made rather than at the till.	Critical
FE-4	Automatic shopping-list generation	Derive the list as the difference between what the plan requires and what the pantry holds, aggregating quantities across recipes and grouping items for a single pass through a store. Items checked off while shopping return to the pantry, closing the loop without a second round of data entry.	Critical
FE-5	Recipe catalogue, search, and pantry-aware suggestion	Browse and search easy recipes by preparation time, cost, cuisine, and dietary compatibility; and, beyond search, receive suggestions ranked by how much of the required ingredients the student already owns, weighted toward near-expiry items and constrained by profile and budget. This	Critical

ID	Feature	Description	Priority
		answers the question students actually ask — not “what is a good recipe” but “what can I cook tonight.”	
FE-6	Waste reduction and expiry awareness	Track expiry dates, surface items about to spoil, prompt with recipes that consume them first, and show a running indication of food and money saved.	Important
FE-7	Nutrition summary	Present nutrition information per recipe, meal, and day, and the shape of the week against stated goals, in plain non-clinical language with an explicit non-medical-advice statement. The purpose is to make an unbalanced week visible while it can still be changed.	Important
FE-8	Content moderation workspace	Work through a queue of submitted and imported recipes: approve, edit, or reject; correct ingredient, allergen, cost, and nutrition metadata; retire unsafe published content; review an audit trail. Filters and batch actions are required because throughput is the constraint on this role. No recipe reaches a student without passing this workflow, which is what makes FE-1’s guarantee meaningful.	Critical
FE-9	System administration and monitoring	Manage accounts and roles, monitor availability, performance, and error rates, review security and audit logs, and perform backup and restore. Invisible to the student and still required in the first release: the system holds health-adjacent data, and losing a student’s pantry and plans destroys the product’s value to them.	Critical
FE-10	Grievance and support handling	Raise a grievance from inside the application; triage, respond to, and resolve tickets with a permissioned and logged view of the affected account; escalate confirmed defects and feed recurring issues back into the backlog.	Important
FE-11	Feedback and recipe rating	Rate a recipe, see aggregate ratings, and report an inaccuracy such as a missing allergen or a wrong quantity. Reports route directly into the moderation queue, closing the loop between the student community and content quality.	Useful

6. Other Product Requirements

Standards, platform, and environment

- Standards: WCAG 2.1 Level AA for all student-facing pages; Quebec Law 25 and GDPR where applicable (consent, minimisation, access logging, right to deletion) for personal data including allergens; OWASP Top 10 as the security baseline; national food-labelling reference quantities for nutrition figures.
- Platform: a web-based application connected to a relational database on commodity cloud infrastructure. No installed client for any user class. Clients: the two most recent major versions of mainstream desktop and mobile browsers, JavaScript enabled, no plug-ins, no specialised hardware. External interfaces: an email service and an optional public nutrition dataset.
- Environment: students use the system at home, in the library, and in grocery stores, on laptops, shared machines, and phones, under variable connectivity; the shopping list must be legible and operable one-handed in a store aisle. Back-office users work from desktop machines on stable connections. The system must be operable by a small team — roughly one administrator, one or two moderators, one support specialist.

Quality ranges

Attribute	Range	Priority
Performance	Pages respond within 2 seconds; recipe suggestion within 3 seconds; shopping-list generation within 2 seconds — at 200 concurrent sessions.	High
Scalability	At least 5,000 registered students and 200 concurrent sessions in the first release.	Medium
Availability	99% monthly. Planned maintenance outside 17:00–21:00 local time, the evening planning peak.	High
Usability	A first-time student, untrained, adds a pantry item in under 15 seconds and produces a first weekly plan in under 10 minutes.	High
Robustness	The shopping list remains usable on a degraded connection. Failure of the nutrition dataset degrades FE-7 only; planning, suggestion, and list generation continue.	High
Content correctness	Zero published recipes with incorrect allergen labelling; every recipe passes moderation before publication. An allergen error is a safety incident, not a content defect.	High
Security	Credentials hashed; TLS throughout; role-based access control; every Support access to student data is logged.	High
Recoverability	Daily backups; recovery point objective 24 hours, recovery time objective 4 hours.	High
Supportability	Support tickets acknowledged within one business day; moderation queue cleared within two.	Medium

Constraints

- The product must be web-based and connected to a database; no native or installed client in this release.
- It must run on commodity cloud infrastructure within the operating budget of a free student service.
- Nutrition information must never be presented as medical or dietetic advice, and a disclaimer must appear wherever it is shown.
- Recipe and nutrition content must be licensed to permit redistribution within the platform.
- Advertising and paid placement are excluded from recipe suggestion, since paid ranking would undermine the trust FE-5 depends on.

- Out of scope: grocery-store ordering and payment integration, native mobile applications, wearable-device integration, and clinical dietary prescription.

Documentation requirements

- Student user guide (Medium): a short online guide covering registration, dietary profile, pantry entry, planning, and shopping. It must be short — a long student manual is a symptom of a usability failure, not a remedy.
- Online help (Medium): context-sensitive inline guidance, including why a recipe was excluded by the dietary profile, and a searchable FAQ.
- Operational guides (High): deployment, monitoring, and backup procedures for the Administrator; a moderation-workflow guide for the Moderator; a triage and escalation guide for Support. These users are trained and will read documentation.
- Labelling and packaging (Medium): consistent Healthy Living branding across all role-specific views; a visible non-medical-advice statement on every nutrition view.
- Installation (Low): no end-user installation guide is required; the product runs in a browser.

Priority of these requirements

Content correctness, security, privacy compliance, and the usability of pantry entry are High and non-negotiable in the first release: the first three are the conditions under which the product may be operated at all, and the fourth is the condition under which anyone will keep using it. Performance, availability, and recoverability are High but tunable. Scalability, supportability, and the documentation requirements are Medium and are expected to change as the deployment grows beyond a single campus. The greatest risk among these is content correctness, since it depends on human moderation capacity rather than on code, and its failure is a safety incident rather than an inconvenience.

Appendix A — Effort Log (Task 0)

Date	Section / Activity	Hours	Notes
July 9, 2026	Reading the assignment; studying the RUP Vision template	0.5	Setup
July 9, 2026	Section 1 — Introduction and References	0.75	
July 9, 2026	Section 2 — Positioning	1.0	
July 10, 2026	Section 3.1–3.2 — Stakeholder and user summaries	1.0	
July 10, 2026	Section 3.3 — User environment	0.75	
July 11, 2026	Section 3.4 — Key stakeholder and user needs	1.5	Longest section
July 11, 2026	Section 4 — Product overview, assumptions and dependencies	1.0	
July 12, 2026	Section 5 — Product features	1.5	
July 12, 2026	Section 6 — Other product requirements	1.25	
July 12, 2026	Review, formatting, version control	0.75	Committed to repository
	Total	10.0	

Version control: this document is maintained in a repository so that revisions can be tracked and updated with feedback from future deliveries. Repository: <https://github.com/sachin181/soen6481-healthy-living-vision>